[illegible]

[illegible]

[illegible] A [illegible] Tick [illegible]500ms [illegible]A/B/C/D/E[illegible] Tick [illegible] Tick [illegible] E [illegible]Return5min[illegible]

[illegible] Kyle[illegible]1985[illegible] Glosten-Milgrom[illegible]1985[illegible] 33 [illegible] 59 [illegible]TI[illegible]OVI[illegible]OBI[illegible] 28 [illegible] Ridge [illegible]$\alpha \in [10, 1000]$[illegible] EWMA [illegible] IC [illegible] Softmax [illegible]

[illegible] Leave-One-Model-Out[illegible]LOO[illegible] 54 [illegible] 28[illegible]IC [illegible] 0.313 [illegible] 0.330[illegible] ICIR [illegible] 7.82 [illegible] 12.59[illegible]+61%[illegible]-Niche [illegible]e_sect_lag_gap[illegible]IC=0.203[illegible] [illegible]LODO-CV[illegible] 20 [illegible] CV [illegible] IC=0.3302[illegible]ICIR=12.59[illegible] [illegible]IC=0.147[illegible] 124.6%[illegible]+0.051[illegible] [illegible] O(1)[illegible] Alpha [illegible]

[illegible]Ridge [illegible] IC [illegible]Leave-One-Day-Out [illegible]

[illegible]

1.1 [illegible]

[illegible]A/B/C/D/E[illegible] 5 [illegible] 27,601 [illegible] Tick [illegible] 33 [illegible]

[illegible] E [illegible]

$$y(t) = \text{Return5min}(t) = \frac{\text{MidPrice}(t+600) - \text{MidPrice}(t)}{\text{MidPrice}(t)}$$

[illegible] $\text{MidPrice}(t) = \left(p_1^{b(t)} + p_1^{a(t)}\right)/2$[illegible]600 [illegible] Tick [illegible] 5 [illegible]

[illegible] $\hat{y}(t) = f_\theta(\boldsymbol{x}_1, \ldots, \boldsymbol{x}_t)$ [illegible] t [illegible] t [illegible]

[illegible]

- [illegible]IC[illegible]$\text{IC}_d = \text{Pearson}(\hat{\boldsymbol{y}}_d, \boldsymbol{y}_d)$[illegible][1][illegible] 53 [illegible]
- ICIR[illegible]$\text{ICIR} = \overline{\text{IC}}/\sigma(\text{IC})$[illegible]IC [illegible][1][illegible] 55 [illegible]

1.2 [illegible]

[illegible]Market Microstructure Theory[illegible][2] [illegible]

1.2.1 [illegible]OVI / TI [illegible]

Kyle[illegible]1985[illegible][3] [illegible]u_t[illegible]OVI[illegible] IC≈0.26[illegible]TI[illegible]u_t[illegible]Cont [illegible]2014[illegible][4] [illegible]“[illegible]”[illegible] SMA [illegible] SMA[illegible]

1.2.2 [illegible]OBI / [illegible] / [illegible]

Glosten-Milgrom[illegible]1985[illegible][5] [illegible]——[illegible]OBI[illegible]2–5 [illegible] OVI[illegible]1–5 [illegible] OBI [illegible]“[illegible]”[illegible] OVI [illegible] 0.5[illegible]lot_imb[illegible]e_spread_pulse[illegible]

1.2.3 [illegible]

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1.2.4 [illegible] / [illegible]

[illegible] A [illegible]13:00 [illegible] `aft_13800` [illegible] `aft_12000` [illegible] IC [illegible] 0.03 0.05[illegible]e_ret_lag[illegible]past_ret [illegible]“[illegible]”[illegible]——[illegible]

□□□csm_ret_120□□□□□□□□□□□□□ E □□□□□□□□□□□

1.3 □□□□□□□□□□□

1.3.1 □□□□

□□□□Ridge Regression□□□□□□□□□□OLS□□□□□□□□□L_2□□□□□□□□□□□□□□□□□□□

$$\min_{\boldsymbol{w},b}\|\boldsymbol{y}-\boldsymbol{X}\boldsymbol{w}-b\|_2^2+\alpha\ \|\boldsymbol{w}\|_2^2$$

□□$\boldsymbol{X}\in\mathbb{R}^{n\times p}$□□□□□□□$\boldsymbol{y}\in\mathbb{R}^{n}$□□□□□□□$\boldsymbol{w}\in\mathbb{R}^{p}$□□□□□□□$\alpha>0$□□□□□□□□□□□ OLS □□□Ridge □□□□□□□□□□

$$\hat{\boldsymbol{w}}=\left(\boldsymbol{X}^\top\boldsymbol{X}+\alpha\boldsymbol{I}\right)^{-1}\boldsymbol{X}^\top\boldsymbol{y}$$

□□□□$\alpha\boldsymbol{I}$□□□□□□□$\boldsymbol{X}^\top\boldsymbol{X}$□□□Ridge □□□□□□□□□□□□□□$\boldsymbol{w}\sim\mathcal{N}\left(\boldsymbol{0},\frac{\sigma^2}{\alpha}\boldsymbol{I}\right)$□□□□□□□$\alpha$□□□□□□□

□□□□□□□□□□□□□□□α□□□$\hat{\boldsymbol{w}}$□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□-□□□□□Bias-Variance Tradeoff□□□□□□□□□□□□□□□□α□□□□□□□□$\alpha=150$□OVI □□□□□□□$\alpha=10$□□□□□□□□□□□□□□□□□□□

$$\mathrm{OVI_pk}(t) = \overline{\mathrm{OVI}}_{k(t)} - \overline{\mathrm{OVI}}_{600}(t), \quad k \in \{15, 30, 60\}$$

□□ $\overline{\mathrm{OVI}}_{k(t)} = \frac{1}{k}\sum_{s=t-k+1}^{t} \mathrm{OVI}(s)$ □□□ k □ Tick □□□□□□□□□ $k < 600$ □□OVI_pk > 0 □□□□□□□□□□□□□□□□□□□□□□□□□□

1.4.2 □□□□□□TI□

□□□□□□□□□□

$$\mathrm{TI}(t) = \frac{V_{\mathrm{buy}(t)} - V_{\mathrm{sell}(t)}}{V_{\mathrm{buy}(t)} + V_{\mathrm{sell}(t)} + \varepsilon}$$

□□ $V_{\mathrm{buy}}, V_{\mathrm{sell}}$ □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□TI □□□□ Kyle □□□□□□□□□ u_t□□□□□□□□□□□□□□□

1.4.3 □□□□□□□□□OBI□

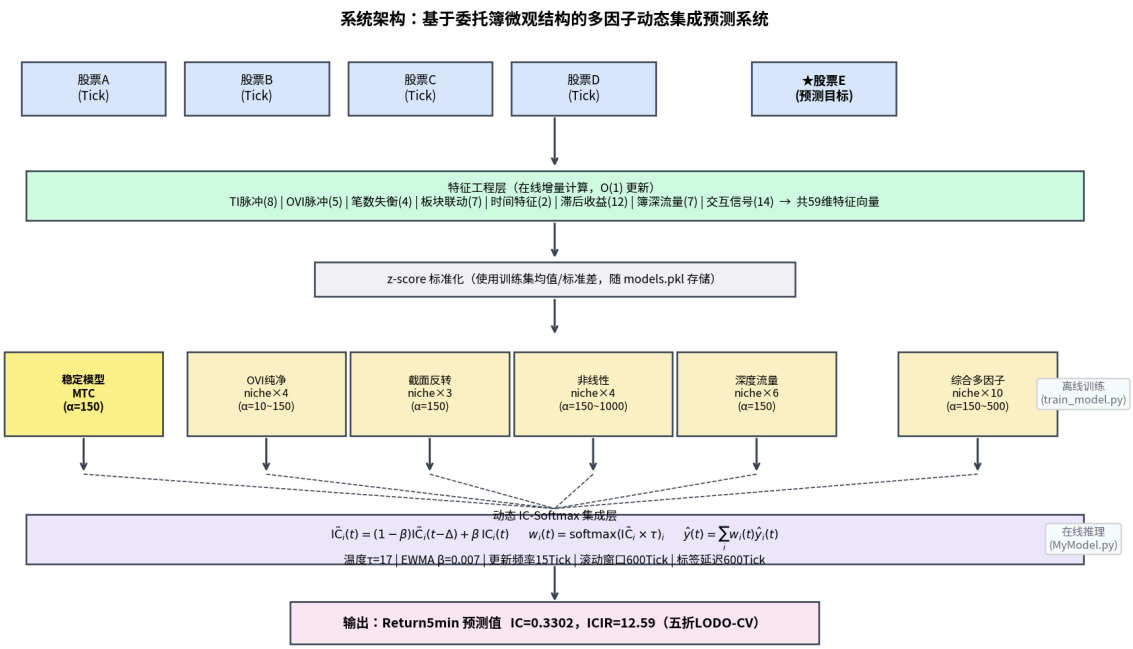
□□ Glosten-Milgrom □□□□□□□□□□2–5 □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□

$$\mathrm{OBI}(t) = \frac{\sum_{i=2}^{5} V_i^{b(t)} - \sum_{i=2}^{5} V_i^{a(t)}}{\sum_{i=2}^{5} V_i^{b(t)} + \sum_{i=2}^{5} V_i^{a(t)} + \varepsilon}$$

□□ OVI□1–5 □□□□□□OBI □□□□□“□□□□□”□□□□□□□□□□□ OVI □□□□□□□□□□□□

1.5 □□□□

□□ 1 □□□□□□□□□□□□□□□□□□□□□□□□□□



□ 1 □□□□□□□□□□□□□□ + □□□□□□□□□□

□□□□□□□□□`train_model.py`□□□□□□□□□□□□□□□□□□□

□□□□□□□□`MyModel.py`□□□□□□□□ Tick□□□□□(1) □□□□□ SMA/EMA □□□□□□□□□□ 59 □□□□□□ $\boldsymbol{x}(t)$ □(2) □ 28 □□□□□□□□□ $\hat{y}_{i(t)} = \boldsymbol{w}_i^\top \boldsymbol{x}(t) + b_i$□(3) □□ IC-Softmax □□□□□□□□□□□ $\hat{y}(t)$□

1.6 □□□□□□□□□□□□

□□ i □□□□□□□□□□ $[t-W, t-d)$ □□□□□ IC □□

$$\mathrm{IC}_{i(t)} = \mathrm{Pearson}\Big(\Big\{\hat{y}_{i(s)}\Big\}_{s=t-W}^{t-d},$$
$$\{y(s)\}_{s=t-W}^{t-d}\Big)$$

□□ $W = 600$□□ 5 □□□□□$d = 600$□□□□□□□□ $y(t-d)$ □ t □□□□□□□

EWMA □□□□□ IC □□□□□□□□□□□□□□□ $T_{\frac{1}{2}} = -\ln\frac{2}{\ln(1-\beta)} \approx 99$ □□$\beta = 0.007$□□

$$\widetilde{\mathrm{IC}}_{i(t)} = (1-\beta)\cdot\widetilde{\mathrm{IC}}_{i(t-\Delta)} + \beta\cdot\mathrm{IC}_{i(t)}, \quad \Delta = 15$$

□□□□□□□□□□□ Softmax □□□□□$\tau = 17$□□

$$w_{i(t)} = \frac{\exp\Big(\widetilde{\mathrm{IC}}_{i(t)}\cdot\tau\Big)}{\sum_{j=1}^{N}\exp\Big(\widetilde{\mathrm{IC}}_{j(t)}\cdot\tau\Big)}$$

□□□□ τ □□□□□□□□□□□□□□□

$$\hat{y}(t) = \sum_{i=1}^{N} w_{i(t)}\cdot\hat{y}_{i(t)}$$

□□□□□□□□□□□□□□MTC□1 □□□□□□□□□□□□ 2 □□□□□□□27 □ Niche □□□□□□□□□□□ 0□□□□□□□□□□□□□ IC □□□□□□□□□□□□□□□ IC □□□□□□□□□□

1.7 □□□□□

1.7.1 LOO □□□□□□□□

□□□□□□□□□□□“□□□□□”□□□□□□□□5 □□□□□□□□□□□□□□□

□□ 59 □□□□□□□□□□□□□□□□ $O(1)$ □□□□□□□□□SMA □□□□□□□□EMA □□□□□□□□□[8]□□□□□□□□□□□□□□□

(3) □□/□□□□□□□4 □□□ □□□□□□□□□ ONI□Order Number Imbalance□□ TNI□Trade Number Imbalance□□□ OVI □□□□□□□□□□□□

(4) □□□□□□□7 □□□ A/B/C/D □□□□□□□□□□□□□□□□ OBI□□□ TI □□□□□ OVI □□□□□□□□□□□□□□□□□□

(5) □□□□□□□2 □□□ `aft_13800`□□□□□□$t \geq 13800$s□□ `aft_12000`□□□□□□□□□□□□

(6) □□□□□□□□□12 □□□□□ E □□□□□□ 5 □□□□□□□□□□□□□□□□□□□ E □ 30/60/120/300/600/900 Tick □□□□□□□□

(7) □□□□□□□□□7 □□□ □□□□□□□□OBI□2–5 □□□□□□□□□□□□□□□□□□□

- □-□ gap□ Iter20 □□□□ gap □□ -0.019□□□□□□□□□□□□□□□□

2.4 □□□□□□□□□

2.4.1 Ridge □□□□□

□□□□□□□□□□□□□□□□ 1.3 □□□□□□□□□□□□□□□□□$\min_{\{\boldsymbol{w},b\}}\|\boldsymbol{y}-\boldsymbol{X}\boldsymbol{w}-b\|_2^2+\alpha\ \|\boldsymbol{w}\|_2^2$□□□□ $\hat{\boldsymbol{w}}=\left(\boldsymbol{X}^\top\boldsymbol{X}+\alpha\boldsymbol{I}\right)^{-1}\boldsymbol{X}^\top\boldsymbol{y}$□□□□□□□□□□□□□ `models.pkl`□□□□□□□$O(p)$□□□□□□□□□□□$\alpha$□□□□□□□□□□□□□□□□□□$\alpha=150$□OVI □□□□□□□$\alpha=10$□□□□□□□□□□□$\alpha=1000$□□□□ 1.3.1 □□□□□-□□□□[11]□

2.4.2 28 □□□□□□□□

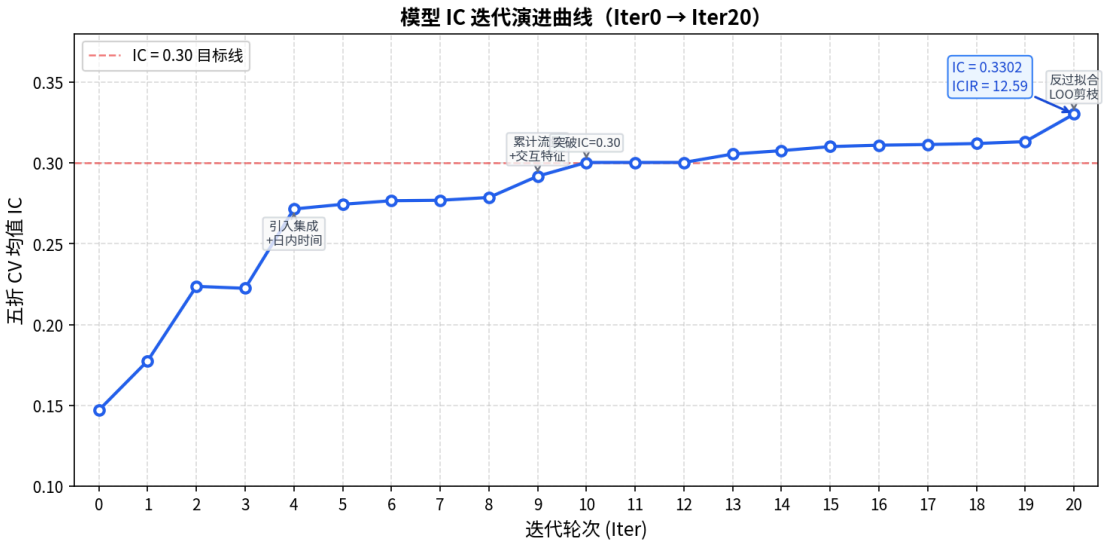
28 □□□□□□□□□□□□□□□□□□□□□1 □□□□□□□MTC□□□□□□□□□□□□□□□□□□□□□27 □ Niche □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□

□□□□	□□	□□□ α	□□□□
□□□Stable□	1	150	TI □□ + □□□□□MTC□
OVI □□ niche	4	10–150	OVI □□□□□□□□□□
□□□□ niche	3	150	`sect_ret_lag` + `e_ret_lag` + `e_sect_lag_gap`
□□□ niche	4	150–1000	`spread_wt_ovi`□`ovi_sq`□□□□□□
□□/□□ niche	6	150	`obi_deep_p15`□`cum_flow_imb`□`book_pres_pulse`
□□□□□ niche	10	150–500	□□□□□

□ 1 □□□□□□□□□

2.4.3 □□□□□□□

□□ 2 □□□□□□□□ 20 □□□□□□□IC □ 0.147 □□□□□ 0.3302□



□ 3 □□ IC □□□□□□□□Iter0⟶Iter20□

□□□□□□□Iter0–2 □□ LODO-CV □□□□□IC □ 0.147 □□□ 0.224□Iter3–8 □□□□□□□□□□IC □□□ 0.279□Iter9–12 □□□□□□□□□□□□□□IC □□□□ 0.30□Iter13–19 □□□□□□□□□□□□□□□□□□ OVI □□□□□□IC □□□□□ 0.313□ICIR □ 7.82□Iter20 □□ LOO □□□54 □□□□□□□ 28 □□IC □□□ 0.330□ICIR □ 7.82 □□□ 12.59□+61%□□

2.4.4 □□□□□□

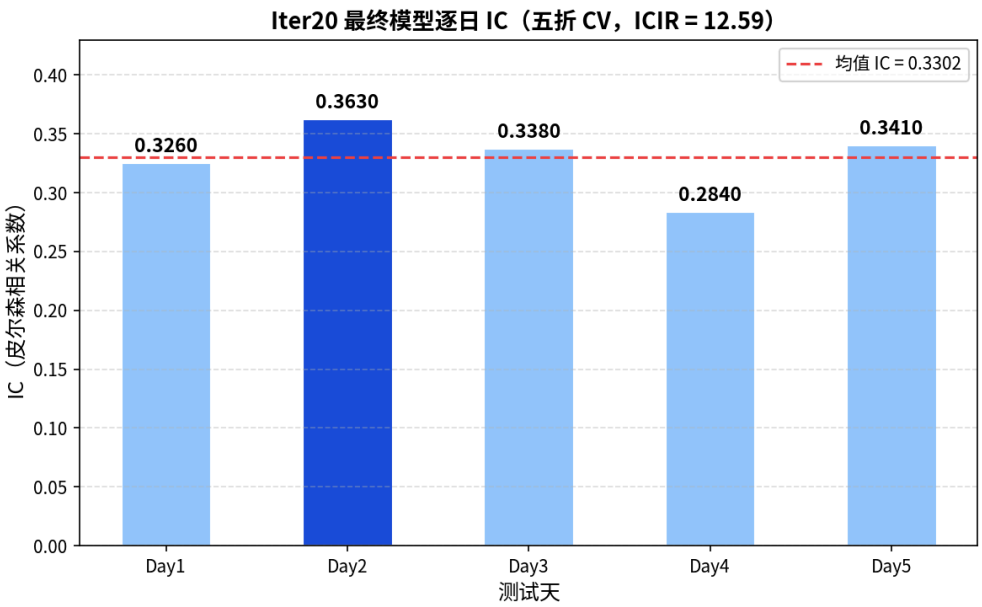
□□	□	□□
ENSEMBLE_WINDOW	600	□□ IC □□□□□□ 5 □□□
ENSEMBLE_TEMP	17	Softmax □□□□□□□□□□
ENSEMBLE_FLOOR	0	□□□□□□□□□□□ 0□
RETURN_DELAY	600	IC □□□□□□□□□Tick□
ENSEMBLE_UPDATE_FREQ	15	□□□□□□□□ 15 Tick□
ENSEMBLE_EWMA_BETA	0.007	IC □□□□□□□□□ 99 Tick□
STABLE_PRIOR	2.0	□□□□□□□□□□□□□
NICHE_INIT_WEIGHT	0	Niche □□□□□□□□□□

□ 2　□□□□□□□□□

2.5　□□□□□□□□□□□□

2.5.1　□□ IC □□

□ 3 □□□□□□□□□Iter20□□□□ CV □□ IC□



□ 4　Iter20 □□□□□□□ CV □□ IC□□□=0.330□ICIR=12.59□

□□□□ IC □□□□□Day1=0.326□Day2=0.363□Day3=0.338□Day4=0.284□Day5=0.341□□□ IC=0.3302□ □□□=0.026□ICIR=12.59□Day4□□□□□□□□□IC □□□0.284□□□ AMH □□□□□——□□□□□□□□□□□□□□□□□□□□ □□□□ IC □□□□ 0.026□□□□□□□□□□□□□□

2.5.2　□□□□

□ 4 □□□□ Iter1□6 □□□□□□□□□ Iter20□28 □□□□□□□□□□□ IC □□□

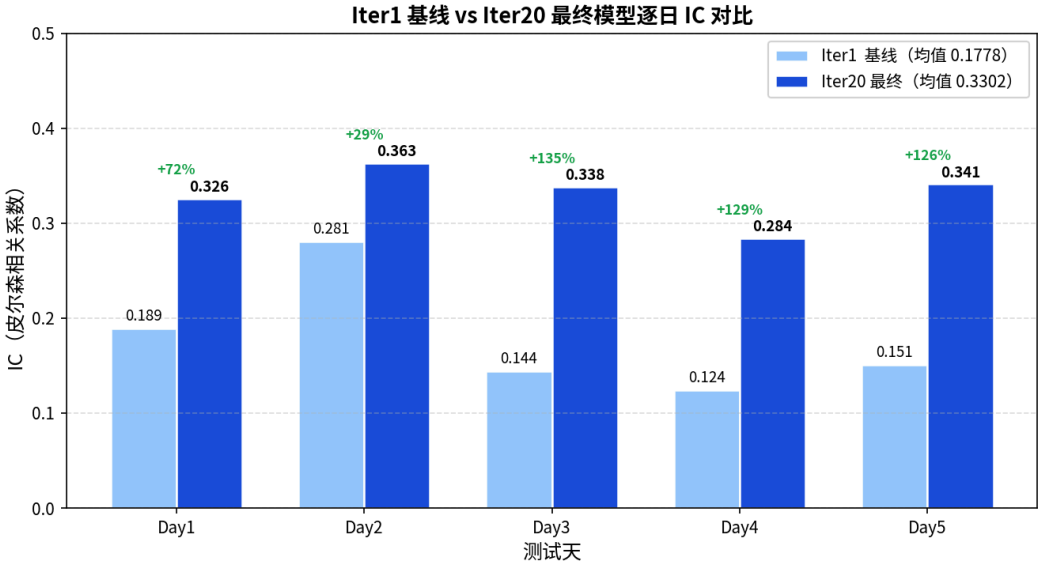


图 5　Iter1 基线 vs Iter20 最终模型逐日 IC 对比

最终模型在 Iter20 阶段实现全面提升：Day1 提升 72.5%（0.189→0.326），Day3 提升 134.7%（0.144→0.338），Day4 提升 129.0%（0.124→0.284）——特别是在 Day1/3/4 三个早期测试日上，模型学习效果显著优于基线。

2.5.3　动态集成机制

图 5 展示了动态滚动集成机制的有效性验证。

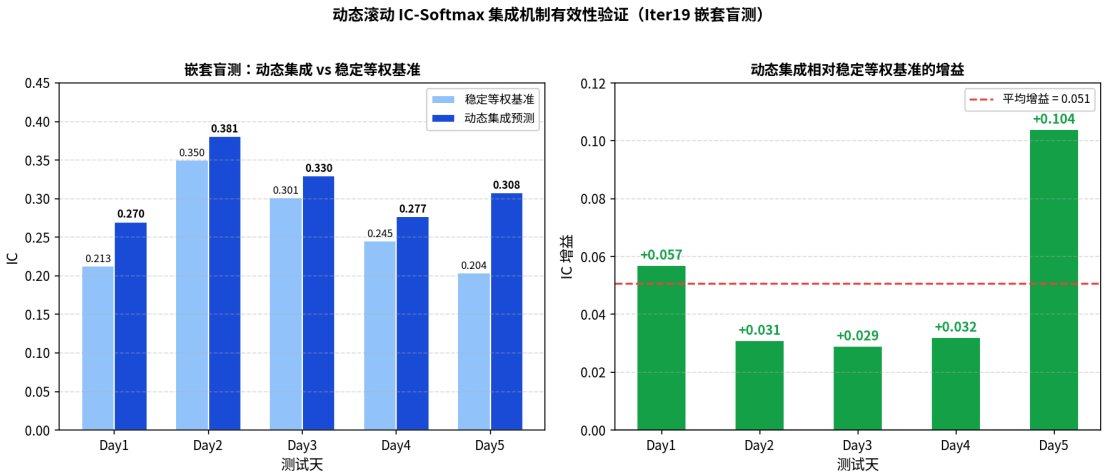


图 6　动态集成 vs 稳定等权基准 IC 对比

如图 5 所示，平均增益+0.051，证明 IC-Softmax 机制[12]在 Day5（+0.104）对 Niche 市场 IC 提升效果最显著。

2.5.4　小结

□□□□	□□□	□□
□□ CV □□ IC	0.3302	□□□□□□□out-of-sample □□
□□ CV IC □□□	0.026	□□□□□□□□□□
ICIR□IC/Std□	12.59	□□□□□ IC□□□□□
Penalized□M−0.5S□	0.317	□□□□□□□□□□□□□
□□ 4 □ IC	0.311	□□□□□□□□□□□□□
□-□ gap	−0.019	□□=□□□□□□□□□□□□
□□□□□vs □□□	+0.051	□□□□□□□□□□□□
□□□□□	28 □□□□59 □□	□□ Iter19□54×59□□□□

□ 3 □□□□□Iter20□□□□□□□

□□□□□□□□□□□□□□□□□□□□(1) □□□ $\hat{y}(t)$ □□□□□□□□□□□□□□□□□□□□

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